f (kHz)	U_R / Vpp (V)	U / Vpp (V)	U_R/U	φ_R (rad)
0.3	0.364	2.04	0.1784	1.4175
0.6	0.724	2.04	0.3549	1.2083
0.9	0.984	2.04	0.4824	1.0467
1.2	1.176	2.04	0.5765	0.9297
1.5	1.384	2.04	0.6784	0.8068
1.6	1.380	2	0.6900	0.8304
1.7	1.436	2	0.7180	0.7562
1.8	1.508	2.04	0.7392	0.7295
2	1.564	2	0.7820	0.6698
2.4	1.684	2	0.8420	0.6122
3	1.812	2	0.9060	0.5297
3.6	1.808	2	0.9040	0.4501
4.2	1.884	2	0.9420	0.3721
5	1.876	1.96	0.9571	0.2827

f (kHz)	U_C / Vpp (V)	U / Vpp (V)	U_C/U	φ_C (rad)
0.3	1.968	2.04	0.9647	-0.1813
0.6	1.880	2.04	0.9216	-0.3404
0.9	1.724	2	0.8620	-0.5055
1.2	1.636	2	0.8180	-0.6100
1.5	1.432	2	0.7160	-0.7342
1.6	1.400	1.96	0.7143	-0.8002
1.7	1.368	2	0.6840	-0.8364
2	1.260	2	0.6300	-0.8872
3	0.988	2	0.4940	-1.0801
4.2	0.720	2	0.3600	-1.1822
5	0.612	1.96	0.3122	-1.2252

f (kHz)	U_R / Vpp (V)	U / Vpp (V)	U_R/U	φ_R (rad)
0.1	1.760	1.9	0.9263	0.0183
1	1.840	1.92	0.9583	-0.0898
2	1.760	1.92	0.9167	-0.1118
4	1.852	1.98	0.9354	-0.2036
7	1.708	1.96	0.8714	-0.3695
11	1.600	1.96	0.8163	-0.5598
15	1.396	1.96	0.7122	-0.7351
17	1.324	2	0.6620	-0.7477
22	1.116	1.96	0.5694	-0.8985
30	0.892	1.88	0.4745	-1.0556
40	0.696	1.8	0.3867	-1.1812

f (kHz)	U_L / Vpp (V)	U / Vpp (V)	U_L/U	φ_L (rad)
0.1	0.112	1.96	0.0571	0.0704

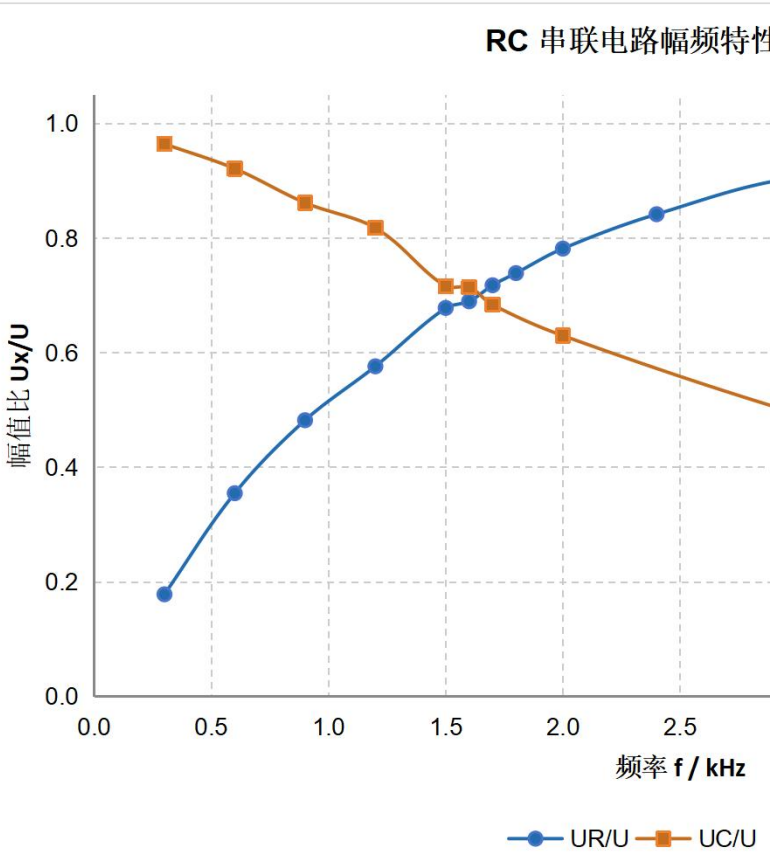
0.5	0.124	1.96	0.0633	0.4514
2	0.244	1.94	0.1258	0.9764
4	0.436	1.94	0.2247	1.1285
7	0.716	1.94	0.3691	1.0248
11	1.024	1.96	0.5224	0.8985
15	1.248	1.96	0.6367	0.7823
17	1.348	1.94	0.6948	0.7263
22	1.568	2	0.7840	0.6220
30	1.764	2	0.8820	0.4712
40	1.840	2	0.9200	0.3770

生成说明：以下为带读数扰动的模型参考值；原表保存在“原表备份”。
原频率和逐点U保留；幅值扰动±3.5%，相位扰动±0.035 rad，固定种子20261029。
电压量化步长0.004 V，时差0.1 μs。模型参数按约3%的交点偏差设定，不代表实测元件。

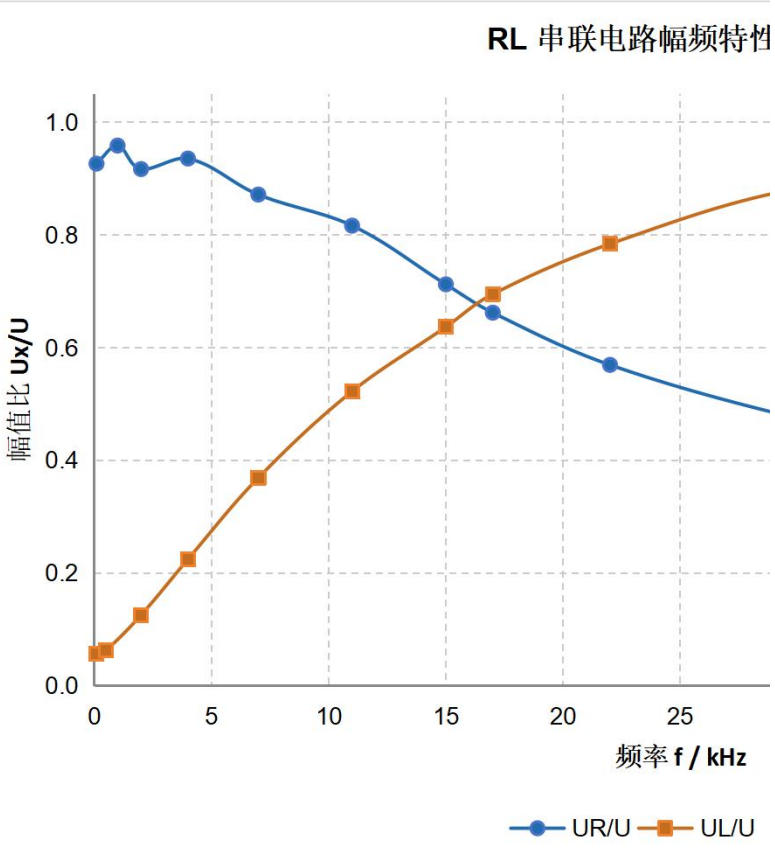
模型参数	数值	单位
电容C	48.538	nF
电感L	9.724	mH

参考曲线	交点/kHz	理论/kHz	相对偏差
RC	1.6417	1.6931	3.04%
RL	16.3937	15.9155	3.00%

ΔT (ms)	T (ms)
0.7520	3.3333
0.3205	1.6667
0.1851	1.1111
0.1233	0.8333
0.0856	0.6667
0.0826	0.6250
0.0708	0.5882
0.0645	0.5556
0.0533	0.5000
0.0406	0.4167
0.0281	0.3333
0.0199	0.2778
0.0141	0.2381
0.0090	0.2000



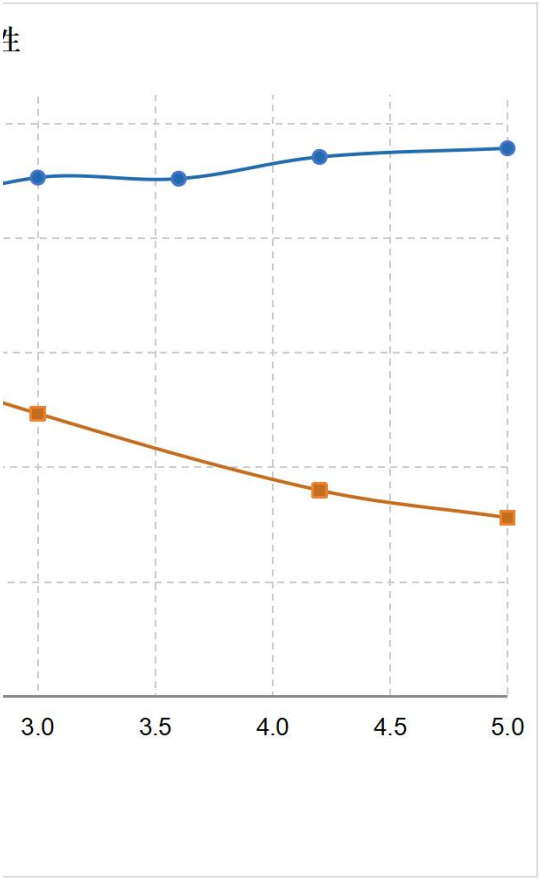
ΔT (ms)	T (ms)
-0.0962	3.3333
-0.0903	1.6667
-0.0894	1.1111
-0.0809	0.8333
-0.0779	0.6667
-0.0796	0.6250
-0.0783	0.5882
-0.0706	0.5000
-0.0573	0.3333
-0.0448	0.2381
-0.0390	0.2000

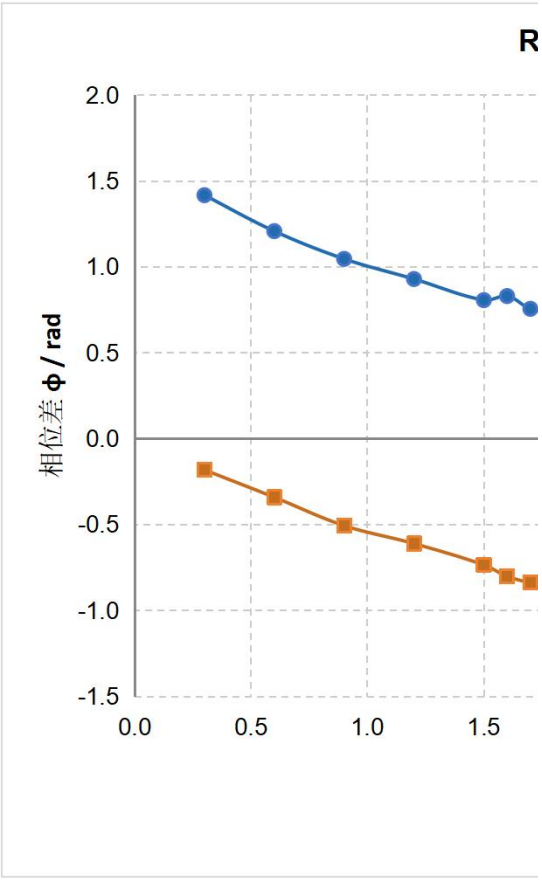


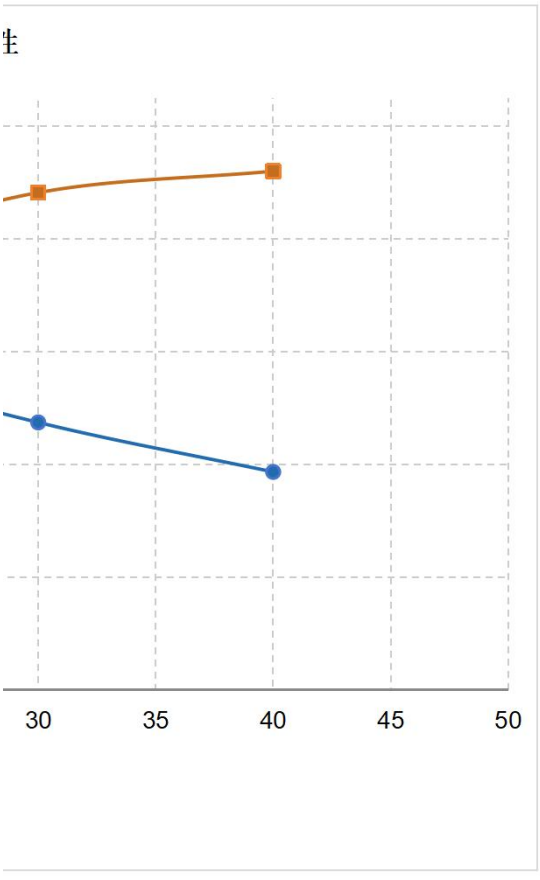
ΔT (ms)	T (ms)
0.0292	10.0000
-0.0143	1.0000
-0.0089	0.5000
-0.0081	0.2500
-0.0084	0.1429
-0.0081	0.0909
-0.0078	0.0667
-0.0070	0.0588
-0.0065	0.0455
-0.0056	0.0333
-0.0047	0.0250

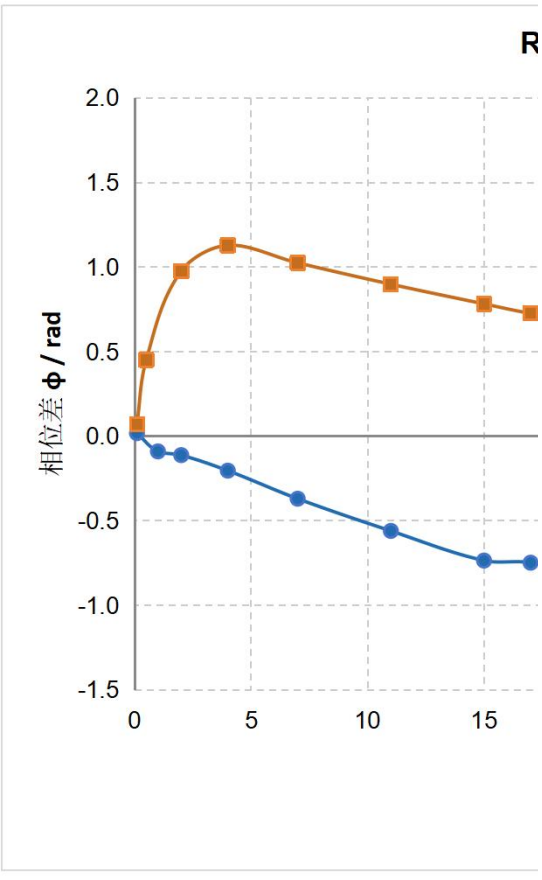
ΔT (ms)	T (ms)
0.1120	10.0000

0.1437	2.0000
0.0777	0.5000
0.0449	0.2500
0.0233	0.1429
0.0130	0.0909
0.0083	0.0667
0.0068	0.0588
0.0045	0.0455
0.0025	0.0333
0.0015	0.0250

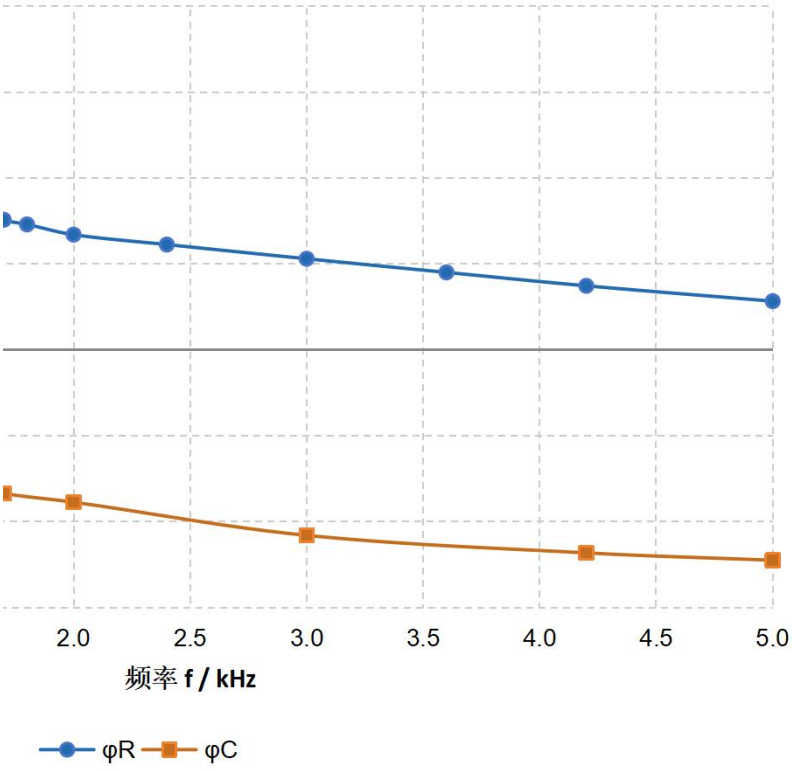

生
3.0
3.5
4.0
4.5
5.0


R
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
相位差 ϕ / rad
0.0
0.5
1.0
1.5

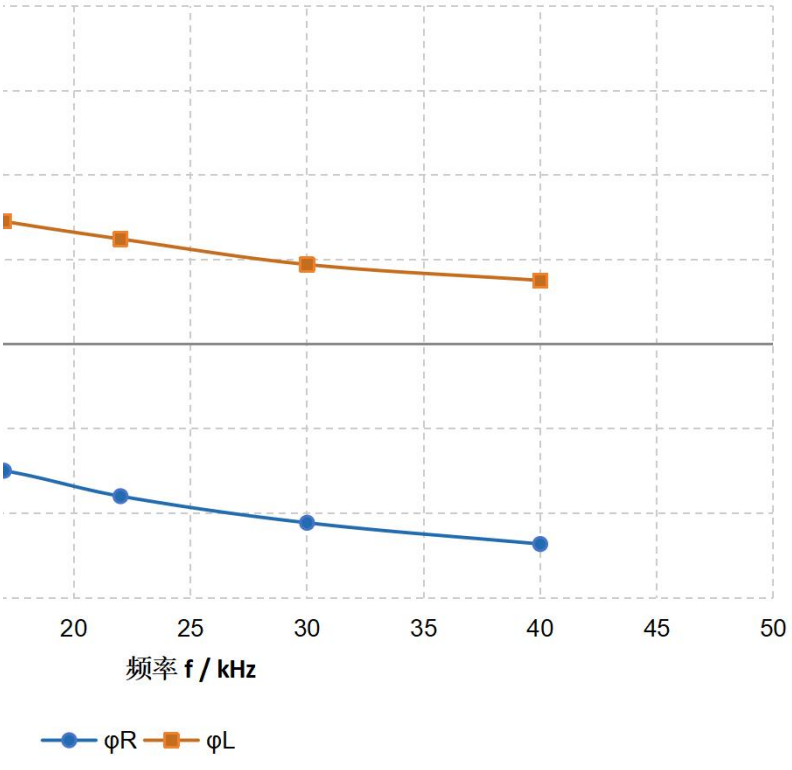

生
30
35
40
45
50


R
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
相位差 ϕ / rad
0
5
10
15

RC 串联电路相频特性



RL 串联电路相频特性



f / kHz	Ux / Vpp	U / Vpp	Ux/U	φ / rad	Δt / µs
0.3	0.356	2.04	0.1745	1.3975	741.4
0.6	0.683	2.04	0.3348	1.2294	326.1
0.9	0.958	2.04	0.4696	1.0806	191.1
1.2	1.177	2.04	0.577	0.9538	126.5
1.5	1.353	2.04	0.6632	0.8445	89.6
1.6	1.372	2	0.686	0.8163	81.2
1.7	1.417	2	0.7085	0.7829	73.3
1.8	1.486	2.04	0.7284	0.7544	66.7
2	1.528	2	0.764	0.7012	55.8
2.4	1.637	2	0.8185	0.6152	40.8
3	1.747	2	0.8735	0.5165	27.4
3.6	1.809	2	0.9045	0.4411	19.5
4.2	1.857	2	0.9285	0.3826	14.5
5	1.858	1.96	0.948	0.3236	10.3

f / kHz	Ux / Vpp	U / Vpp	Ux/U	φ / rad	Δt / µs
0.3	2.005	2.04	0.9828	-0.176	-93.1
0.6	1.92	2.04	0.9412	-0.34	-90.1
0.9	1.763	2	0.8815	-0.489	-86.4
1.2	1.634	2	0.817	-0.615	-81.6
1.5	1.493	2	0.7465	-0.725	-76.9
1.6	1.424	1.96	0.7265	-0.759	-75.5
1.7	1.41	2	0.705	-0.789	-73.9
2	1.295	2	0.6475	-0.87	-69.2
3	0.983	2	0.4915	-1.056	-56
4.2	0.749	2	0.3745	-1.19	-45.1
5	0.627	1.96	0.3199	-1.244	-39.6

f / kHz	Ux / Vpp	U / Vpp	Ux/U	φ / rad	Δt / µs
0.1	1.795	1.9	0.9447	-0.004	-6.7
1	1.812	1.92	0.9438	-0.058	-9.3
2	1.796	1.92	0.9354	-0.116	-9.2
4	1.818	1.98	0.8982	-0.234	-9.3
7	1.708	1.96	0.8714	-0.396	-9
11	1.547	1.96	0.7893	-0.581	-8.4
15	1.379	1.96	0.7236	-0.726	-7.7
17	1.326	2	0.663	-0.79	-7.4
22	1.125	1.96	0.574	-0.912	-6.6
30	0.87	1.88	0.4628	-1.056	-5.6
40	0.662	1.8	0.3678	-1.181	-4.7

f / kHz	Ux / Vpp	U / Vpp	Ux/U	φ / rad	Δt / µs
0.1	0.112	1.96	0.0571	0.0997	158.6

0.5	0.125	1.96	0.0638	0.4543	144.6
2	0.253	1.94	0.1304	1.0053	80
4	0.46	1.94	0.2371	1.1058	44
7	0.752	1.94	0.3876	1.0424	23.7
11	1.075	1.96	0.5485	0.9054	13.1
15	1.303	1.96	0.6648	0.7823	8.3
17	1.378	1.94	0.7103	0.7263	6.8
22	1.59	2	0.795	0.6082	4.4
30	1.74	2	0.87	0.4901	2.6
40	1.844	2	0.922	0.377	1.5